

SADIA ARMIN DIBA

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EDUCATION

Bangladesh University of Engineering and Technology (BUET) **Nov 2022 – Expected July/August 2027**
Bachelor of Science in Biomedical Engineering Dhaka, Bangladesh
CGPA: 3.86/4.00

RESEARCH INTEREST

- ▶ Cardiovascular Mechanics & Hemodynamics
- ▶ Medical Image Segmentation
- ▶ Plaque Rupture Risk Analysis
- ▶ Physics-Informed Neural Networks (PINNs)
- ▶ Computational Fluid Dynamics (CFD)
- ▶ Surrogate Biomechanical Modeling
- ▶ Lesion Aware Learning

RESEARCH EXPERIENCE

Undergraduate Thesis July 2026 – Ongoing	Patient-Specific CFD-Based Hemodynamic Analysis and Machine Learning Prediction of Coronary Artery Flow Patterns Associated with Plaque Formation and Rupture Risk ▶ Investigating hemodynamic predictors of plaque rupture risk, with a broader focus on plaque formation, to inform patient-specific cardiovascular risk stratification ▶ Conducting patient-specific CFD simulations on 3D-reconstructed coronary CTA geometries to quantify hemodynamic parameters (WSS, OSI, pressure drop) linked to atherogenesis ▶ Engineering an automated medical-image-to-CFD segmentation pipeline using Materialise Mimics, 3-matic, and ANSYS Fluent ▶ Synthesizing coronary geometry variations using MorphMan to generate robust synthetic fluid dynamics datasets for model training ▶ Developing a deep learning and Physics-Informed Neural Network (PINN) framework to predict 3D volumetric flow fields and shear stress distributions
Research Intern June 2026 (2 Weeks)	Bangladesh Council of Scientific and Industrial Research (BCSIR), Dhaka, Bangladesh ▶ Gained practical laboratory exposure to analytical workflows, observing instrumentation including HPLC, UPLC, GC, Karl Fischer titration, DSC, and TGA ▶ Observed molecular-biology protocols and equipment including PCR, DNA sequencing, biosafety cabinets, fermenters, incubators, rotary evaporators, and freeze dryers

ACADEMIC PROJECTS

Academic Project Dec 2025 – May 2026	Cardiovascular Baroreflex System Simulation [Github] ▶ Developed and validated a cardiovascular baroreflex model to evaluate Starling-curve responses under varying carotid sinus pressures and simulate acute haemorrhage (20% blood loss) ▶ Implemented a 4-subsystem Simulink model based on Ursino (1999) formulations, solving non-linear differential equations for cardiac output, systemic circulation, and baroreceptor feedback loops
Academic Project Aug 2024 – Jan 2025	Musculoskeletal X-ray Abnormality Detection ▶ Trained DenseNet121, InceptionResNetV2, and EfficientNet architectures for multi-region X-ray abnormality classification across upper limb studies on the MURA dataset (70–80% accuracy)

Academic Project **Respiratory Rate Estimation from PPG Signals** [[Github](#)]

Dec 2025 – May 2026

- ▶ Extracted respiratory rate from noisy PPG signals using Kaiser-window filtering, Hilbert transform, and spectral density estimation on the BIDMC dataset
- ▶ Achieved MAE of 0.88 BPM, RMSE of 1.60 BPM, $r = 0.86$, and 86.8% clinical accuracy within a ± 2 BPM error margin

CAPSTONE PROJECT

Capstone Project **NeoNest – Smart Infant Thermal Management System**

April 2025 – April 2026

- ▶ Developed an Arduino UNO-based prototype integrating DS18B20 temperature sensors, MAX30102, MPU6050, OLED display, and HC-05 Bluetooth for closed-loop thermal monitoring
- ▶ Built safety alert logic and evaluated heating-pad thermal distribution using multi-sensor arrays and thermal imaging

ACHIEVEMENTS & AWARDS

- ▶ **3rd Place** – Project Presentation Competition, 10th Anniversary Celebration, Department of Biomedical Engineering, BUET (Dec 2025)
- ▶ **Dean's List Award**, BUET – Level 1, 2023 – Recognized for outstanding academic performance

TECHNICAL SKILLS

Simulation & Modeling: ANSYS Fluent, MATLAB/Simulink, CFD, Materialise Mimics, 3-matic, MorphMan

Machine Learning & Computing: Deep Learning (DenseNet, InceptionResNet, EfficientNet), Physics-Informed Neural Networks (PINNs), Python, MATLAB

Signal Processing: MATLAB Signal Processing Toolbox – filtering, FFT/DFT, spectral analysis, Hilbert transform

Analytical & Laboratory: HPLC, UPLC, GC, Karl Fischer Titration, DSC, TGA, PCR, DNA Sequencing

Hardware & Embedded: Arduino, ATmega32, DS18B20, MAX30102, MPU6050

RELEVANT COURSEWORK

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|----------------|--------------------------------|-----------------------------------|
| ▶ Anatomy | ▶ Bio-Fluid Mechanics | ▶ Digital Signal Processing |
| ▶ Physiology | ▶ Medical Imaging Systems | ▶ Machine Learning for Healthcare |
| ▶ Biomechanics | ▶ Biomedical Signal Processing | |